

Karen Xiao

Incoming CS PhD Student at UC San Diego

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EDUCATION

University of California, San Diego

Ph.D. Computer Science

Advisors: Leo Porter, Gerald Soosairaj, Bill Griswold

La Jolla, CA

2026-2031

Wellesley College

B.A. Computer Science and Mathematics, *Summa Cum Laude*

Departmental Honors in Computer Science (Honors Thesis)

Wellesley, MA

2023-2026

University of Maryland, College Park

Honors College, Computer Science, GPA: 3.97/4.0

Gemstone Research Program

College Park, MD

2022 –2023

RELEVANT COURSES

Human Computer Interaction; Natural Language Processing; Fundamental Algorithms; Deep Learning; Introduction to Machine Learning; Regression Analysis & Statistical Models; Theory of Computation; Computer Systems; Data Structures; Combinatorics & Graph Theory; Advanced Linear Algebra; Abstract Algebra; Multivariable Calculus; Real Analysis

RESEARCH EXPERIENCE

Senior Honors Thesis

Advisor: Brian Brubach

Wellesley College

August 2025 – present

- Completed a senior honors thesis that proposes methods to measure the diversity of evaluation in classes by treating different types of assignments as alternative ways of ranking students

NSF REU Intern

Human Computer Interaction Institute

Carnegie Mellon University

May 2025 – September 2025

- Worked with Conrad Borchers and Dr. Vincent Aleven in the Creating Adaptive Tutoring Systems (CATS) Lab
- Applied statistical methods to analyze the impact of data-driven redesign on intelligent tutoring systems
- Identified new insights into what constitutes a productive level of difficulty for data-optimized practice

Research Assistant

Institute for Mathematics and Democracy

Wellesley College

December 2024 – September 2025

- Simulated various ranked choice election methods on datasets containing ballots for thousands of real elections
- Conducted empirical comparison of various ranked choice voting methods using Python and VoteKit
- This research is believed to be the largest study of ranked choice elections ever performed

Undergraduate Student Researcher

CrowdNets Lab

Wellesley College

August 2024 - May 2025

- Defined and evaluated a graph-based spatio-temporal POI ranking model that outperforms baseline ML models
- Ran simulations and designed dynamic visualizations on large-scale real-world Uber data to evaluate performance
- Won 2nd place in the Undergraduate Student Research Competition at ACM SIGSPATIAL 2025 (Minneapolis, MN)

Undergraduate Student Researcher

MIT Urban Metabolism Group

MIT

February 2024 - May 2024

- Collected and analyzed data regarding climate risks and policies in various metropolitan clusters
- Converted a policy portfolio into a quantifiable matrix for ML-driven insights

PUBLICATIONS

Liu, Q., Borchers, C., Xia, M., **Xiao, K.**, Carvalho, P., Koedinger, K., & Alevan, V. (2026). Evaluating a Process for the Data-driven Redesign of Educational Technology: Improved Learning Processes. In *The 27th International Conference on AI in Education (AIED '26)*. https://doi.org/10.1007/978-3-032-29760-0_12

Xiao, K., Irisumi, M., & Bassem C. (2025). A Graph-Based Spatio-Temporal POI Ranking Measure for Pickup and Delivery Platforms. In *The 33rd ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '25)*. <https://doi.org/10.1145/3748636.3766535> [SRC 2nd Place Winner]

WORK EXPERIENCE

Financial Data Science Internship

Quantbot Technologies LP

Data Science Intern

May 2024 - August 2024

- Optimized data capture and validation process using DuckDB, Pandas, and Apache Iceberg
- Assessed, enriched, and normalized company outlook and projection data covering all global markets
- Researched new database query tools that the company can implement to optimize data capture toolkits

Financial Data Science Internship

Quantbot Technologies LP

Data Science Intern

May 2023 - August 2023

- Validated and refined raw Equities and Futures data using Python Pandas and NumPy on AWS
- Collaborated with quantitative researchers, traders, and external data vendors
- Optimized existing data processing toolkits using SQLite and Pandas

LEADERSHIP EXPERIENCE

Undergraduate Teaching Assistant

Wellesley College

Wellesley College Computer Science Department

January 2024 - present

- Lead weekly office hours, offer 1-on-1 tutoring, and grade assignments for CS 231 Design & Analysis of Algorithms over the course of 4 semesters

Captain

Wellesley College

Wellesley College Club Squash Team

August 2025 - present

- Lead and organize a team of 15 players by coordinating practice and matches, mentoring new players, and fostering a positive team environment

Networking Chair

Wellesley College

Wellesley Computer Science Club

October 2024 - present

- Manage alumni and recruiting connections to organize and communicate important recruiting, networking, and volunteering events to underrepresented students interested in CS opportunities

AWARDS, FELLOWSHIPS, & GRANTS

Conference Funding

- (2025) **SIGSPATIAL Travel Award (SIGSPATIAL '26) - \$1,000**, National Science Foundation
- (2025) **Science Center Student Conference Travel Grant - \$300**, Wellesley College

Awards

- (2026) **Excellence in Teaching Award in Computer Science**, Wellesley College
- (2026) **NSF GRFP Honorable Mention**, National Science Foundation
- (2025) **Undergraduate SRC 2nd Place Winner**, ACM SIGSPATIAL '26